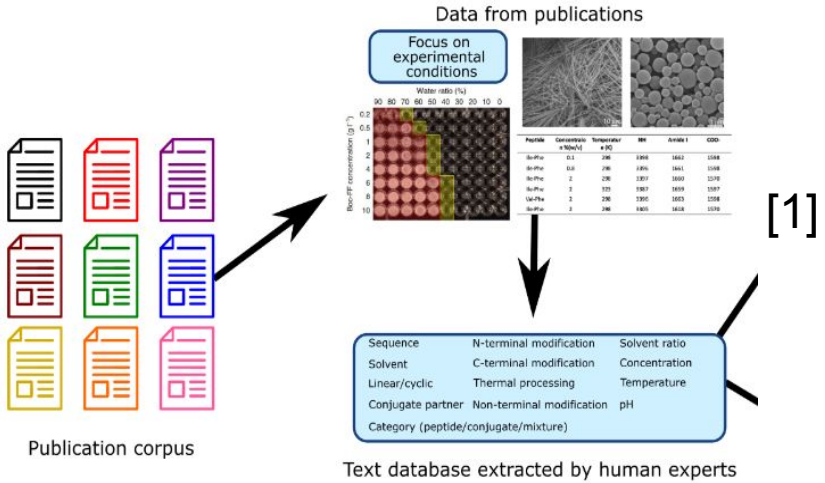
Student Seminar - IDP Presentation 1

Can Ersoy - 22.09.2026

Project Background

- LLM assisted evidence extraction from peptide self-assembly research
- Based on project: https://github.com/lamm-mit/peptideminer [1]

- **Our focus is on 4-20 mer peptides**



[1]

[1]: PeptideMiner: Learning the rules of peptide self-assembly through data mining with large language models, Zhenze Yang and Sarah K. Yorke and Tuomas P. J. Knowles and Markus J. Buehler, 2024

Roadmap

1) Golden Dataset Curation -> SAPdb [2], CPAD 2.0 [3], Cross-Beta [4], Amyload[5]

2) Paper Retrieval Pipeline

3) Extraction Pipeline

[2]: Mathur, Deepika, Harpreet Kaur, Anjali Dhall, Neelam Sharma, and Gajendra P. S. Raghava. “SAPdb: A Database of Short Peptides and the Corresponding Nanostructures Formed by Self-Assembly.” *Computers in Biology and Medicine* 133 (June 2021): 104391. https://doi.org/10.1016/j.compbiomed.2021.104391.
[3]: Rawat, Puneet, R. Prabakaran, R. Sakthivel, A. Mary Thangakani, Sandeep Kumar, and M. Michael Gromiha. “CPAD 2.0: A Repository of Curated Experimental Data on Aggregating Proteins and Peptides.” *Amyloid* 27, no. 2 (2020): 128–33. https://doi.org/10.1080/13506129.2020.1715363.
[4]: Gonay, Valentin, Michael P. Dunne, Javier Caceres-Delpiano, and Andrey V. Kajava. “Developing Machine-learning-based Amyloidogenicity Predictors with Cross-Beta DB.” *Alzheimer’s & Dementia* 21, no. 2 (2025): e14510. https://doi.org/10.1002/alz.14510.
[5]: Wozniak, Pawel P., and Malgorzata Kotulska. “AmyLoad: Website Dedicated to Amyloidogenic Protein Fragments.” *Bioinformatics* 31, no. 20 (2015): 3395–97. https://doi.org/10.1093/bioinformatics/btv375.

Golden Dataset (1)

Columns: **sequence, length, type, n_terminal, c_terminal, temperature, ph, peptide_concentration, solution, solvent, volume_ratio, incubation_period (in min), peptide/conjugate/mixture, conjugate_partner, linear/cyclic, detection_method, preparation_protocol, pmid, database**

- **Currently 1182 entries (1017 from SAPdb/PeptideMiner, 102 from Cross-Beta, 59 from CPAD/Amyload, 4 from AmyLoad)**

- **Pmid per entry to retrieve papers**

Golden Dataset (2)

: PeptideMiner/SapDB

: Cross-Beta DB

: CPAD/Amyload

sequence	length	type	n_terminal	c_terminal	temp	ph	peptide_concentration
FP	2	hydrogel	Chlorination	N-hexadecylamine	25 °C	7	132 mg/ml
ATAVSEWTEYKTADGKTFYYNNRTLE	26	amyloid	Free	Free	37 °C	7.0	100 μM
ISKLEYSNFSVRY	13	amyloid	Free	Free	25 °C	5.5	0.4 mg/mL

Golden Dataset (3)

solution	solvent	volume_ratio	Incubation_ period (min)	peptide/conju gate/mixture	conjugate_p artner	linear/cyclic
H2O	Aqueous dispersion	1	25	Peptide	NP	Linear
Aqueous Buffer	10 mM sodium phosphate, sodium azide 0.02%(v/v)	1	NP	Peptide	NP	Linear
Acetate	50 mM Acetate; TFE (25%)	1	NP	Peptide	NP	Linear

Golden Dataset (4)

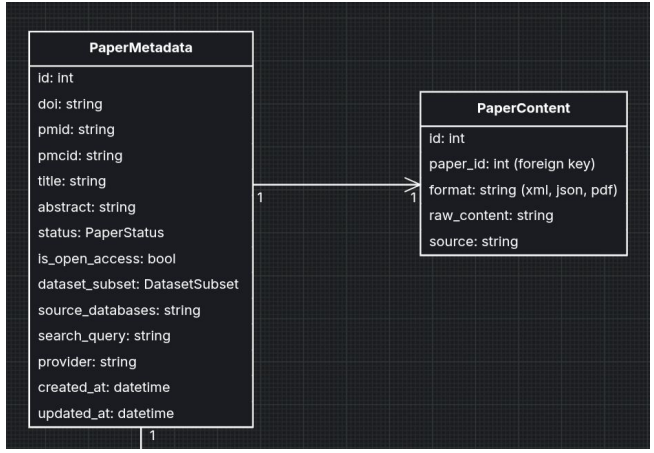
detection_method	preparation_protocol	pmid	database
Field Emission Scanning Electron Microscopy (FE - SEM)	The aqueous dispersions of the required amount of compound was slowly heated to dissolve the compounds in water …	18052148	peptideminer/sapdb
solid-state NMR, EM, ThT fluorescence	Purchased Second WW domain of the human CA150 (Sigma (St. Louis, MO))	17060612	cross-beta
ThT	NP	11799398	CPAD; AmyLoad

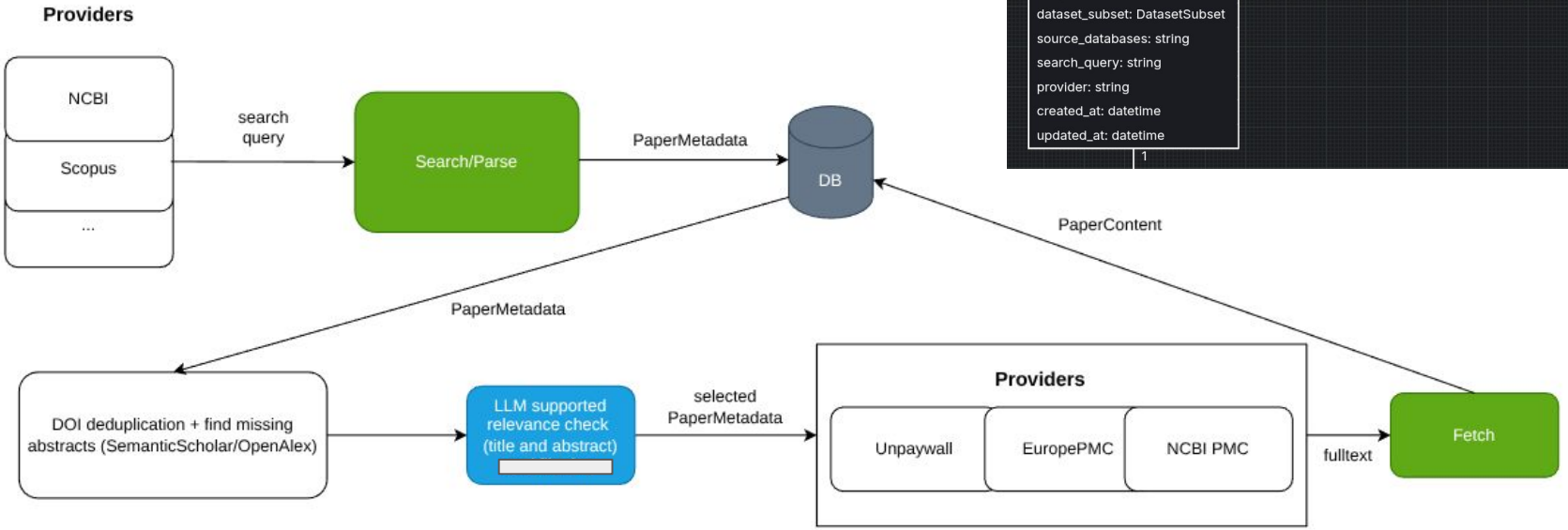
Golden Dataset (5)

Assumptions for Cross-Beta and CPAD databases:

- They can only map column **type** to **amyloid/none**
- **N_terminal** and **C_terminal** are always assumed to be **Free** (?)
- Peptide concentrations are given as is (no normalization to mg/ml)
- Cross-Beta has as **solution** column **'Aqueous Buffer'** if a buffer is present, CPAD has the **buffer name** if present else the same with Cross-Beta
- Both are always assumed to be **Linear**
- Both are always assumed to be **Peptide** (hence no **conjugate_partner**)
- CPAD has for **solvent** column the following concatenation of values: **[Conc] [Buffer]; [Ion Conc] [Ion]; [Additives]**

Paper Retrieval





Paper Search (1)

- Different providers require separate queries but general idea is the same:

NCBI: **(tetrapeptide[tiab] OR pentapeptide[tiab] …) AND (nanostructure[tiab] OR nanotube[tiab] …)**

Scopus: **TITLE-ABS-KEY(tetrapeptide OR pentapeptide …) AND TITLE-ABS-KEY(nanostructure OR nanotube OR nanofiber …)**

Paper Search (2)

Additional requirements for query:

- English only: higher extraction accuracy
- No review article: should increase chance of experiment data being available
- Open access: licensing issues

Results (as of 29.08.2026):

Without add. Requirements:	With add. requirements:
NCBI: 533 papers	187 papers
Scopu: 1608 papers	560 papers

Paper Search (3)

- After doi normalization and deduplication (as of 18.09.2026) there are **574** unique papers from ncbi+scopus.

- **553** have non-empty abstract (after abstract retrieval run)

- **417** also have a pmcid (doesn’t necessarily mean there is a fulltext on pmc but a higher chance)

Paper Relevance Check (1)

SYSTEM_PROMPT = {...} Be careful not to discard papers that do not explicitly state full experimental details in the abstract but hint at experimental work. It is better to include a borderline paper (marking as relevant with lower confidence) than to mistakenly discard a paper containing experimental data.

USER_PROMPT = Here is the title and abstract of a paper:
Title: {title}
Abstract: {abstract}

OUTPUT
detected_clues: JSON-encoded array or text list of detected clues
explanation: Justification focusing on experimental conditions
is_relevant: True if experimental self-assembly data is reported or hinted at
confidence: Confidence score between 0.0 and 1.0

Paper Relevance Check (2)

Golden Dataset Results (116 out of 157 papers tested):

- Gemini 3.5 Flash Lite - 115 relevant - 1 discard
- Gemma 31b - 115 relevant - 1 discard (same paper with gemini 3.5 flash lite)
- Llama 3.2 3b - 112 relevant - 4 discard (also includes same paper as the two above)

System Prompt Specific Section:

You are an expert in scientific paper reading and computational biology.

Your task is to determine whether a given scientific paper is relevant for an information extraction pipeline on peptide self-assembly/amyloid research based only on its title and abstract.

You must determine if the paper reports experimental self-assembly or nanostructure formation of short peptides (e.g., nanofibers, hydrogels, nanotubes, vesicles, organogels, nanospheres, amyloid fibrils) along with experimental conditions (e.g., pH, solvent, concentration, temperature, detection methods).

Paper Relevance Check (3)

Scraped Papers - (100 out of 574) papers - System Prompt v1

- Gemini 3.5 Flash Lite - 78 relevant - 22 discard

System Prompt Specific Section:

You are an expert in scientific paper reading and computational biology.

Your task is to determine whether a given scientific paper is relevant for an information extraction pipeline on 4-20mer peptide self-assembly research based only on its title and abstract.

You must determine if the paper reports experimental self-assembly or nanostructure formation of short peptides (e.g., nanofibers, hydrogels, nanotubes, vesicles, organogels, nanospheres, amyloid fibrils) along with experimental conditions (e.g., pH, solvent, concentration, temperature, detection methods).

Paper Relevance Check (4)

Scraped Papers - (100 out of 574) papers - System Prompt v2

- Gemini 3.5 Flash Lite - 75 relevant - 25 discard

System Prompt Specific Section:

You are an expert in scientific paper reading and peptide research.

Your task is to determine whether a given scientific paper is relevant for an information extraction pipeline on 4-20mer peptide self-assembly/amyloid research based only on its title and abstract. Based on your decision, fulltext of the paper will be retrieved.

You must determine if the fulltext of the paper likely reports experimental self-assembly or nanostructure formation of short peptides (e.g., nanofibers, hydrogels, nanotubes, vesicles, organogels, nanospheres, amyloid fibrils ...) along with experimental conditions (e.g., pH, solvent, concentration, temperature, detection methods ...). Keep in mind, most of the time for longer peptides, oligopeptide naming system (e.g. dodecapeptide for 12-mer) is not provided and peptides might have special names.

Paper Relevance Check (5)

Scraped Papers - (100 out of 574) papers - System Prompt v3

- Gemini 3.5 Flash Lite - 76 relevant - 24 discard

System Prompt Specific Section:

You are an expert in scientific paper reading and biophysics specializing in peptide self-assembly and amyloid aggregation.Your task is to determine whether a given scientific paper is relevant for an information extraction pipeline on 4-20mer peptide self-assembly research based only on its title and abstract. The project extracts experimental records for short peptides (roughly 4-20 residues, including canonical sequences, capped/modified peptides, and amyloidogenic fragments), capturing self-assembly or phase behavior (e.g., nanofibers, hydrogels, nanotubes, nanospheres, amyloid fibrils, ribbons, vesicles, or amorphous aggregates) and associated experimental conditions (e.g., pH, concentration, solvent/buffer, temperature, characterization methods).

Because abstracts are dense summaries that often omit detailed experimental parameters:

- Assume a paper is RELEVANT (`is_relevant: True`) if it reports or strongly implies original experimental (wet-lab) work on peptide self-assembly, aggregation, or amyloid formation.
- Do NOT require full quantitative experimental conditions (such as exact pH or concentration values) or full sequence lengths to be explicitly stated in the abstract; hints of physical characterization (e.g., microscopy, spectroscopy, gelation, kinetic assays) are sufficient.
- Only mark `is_relevant: False` if the paper is unambiguously out of scope: purely theoretical/computational studies without experimental validation, general review articles/editorials, or studies exclusively on intact, full-length proteins without isolated peptide fragments.
- When in doubt, prioritize recall and classify borderline papers as relevant with a lower confidence score.

Paper Fulltext Fetch

- Currently retrieval is only possible from pmc (PubMed Central / NCBI)

- There must be a pmcid provided (can be retrieved via pmid from NCBI API). This doesn’t guarantee full text but is necessary.

Retrieval Counts (as of 20.09)

Golden Dataset: 65/157

Scraped: 385/574

Summary

- Golden dataset includes **1182** entries and **157** unique papers
- Paper search currently supports NCBI and Scopus APIs
- Initial search across NCBI and Scopus returned **574** unique (by doi) papers
- For **553** of these papers, abstract could be retrieved either directly in the initial request or via abstract completion step via OpenAlex/SemanticScholar
- Full Text fetch is currently supported only via PubMed Central (PMC)

Next Steps

- Start with the extraction pipeline
- Improve/complete retrieval alongside